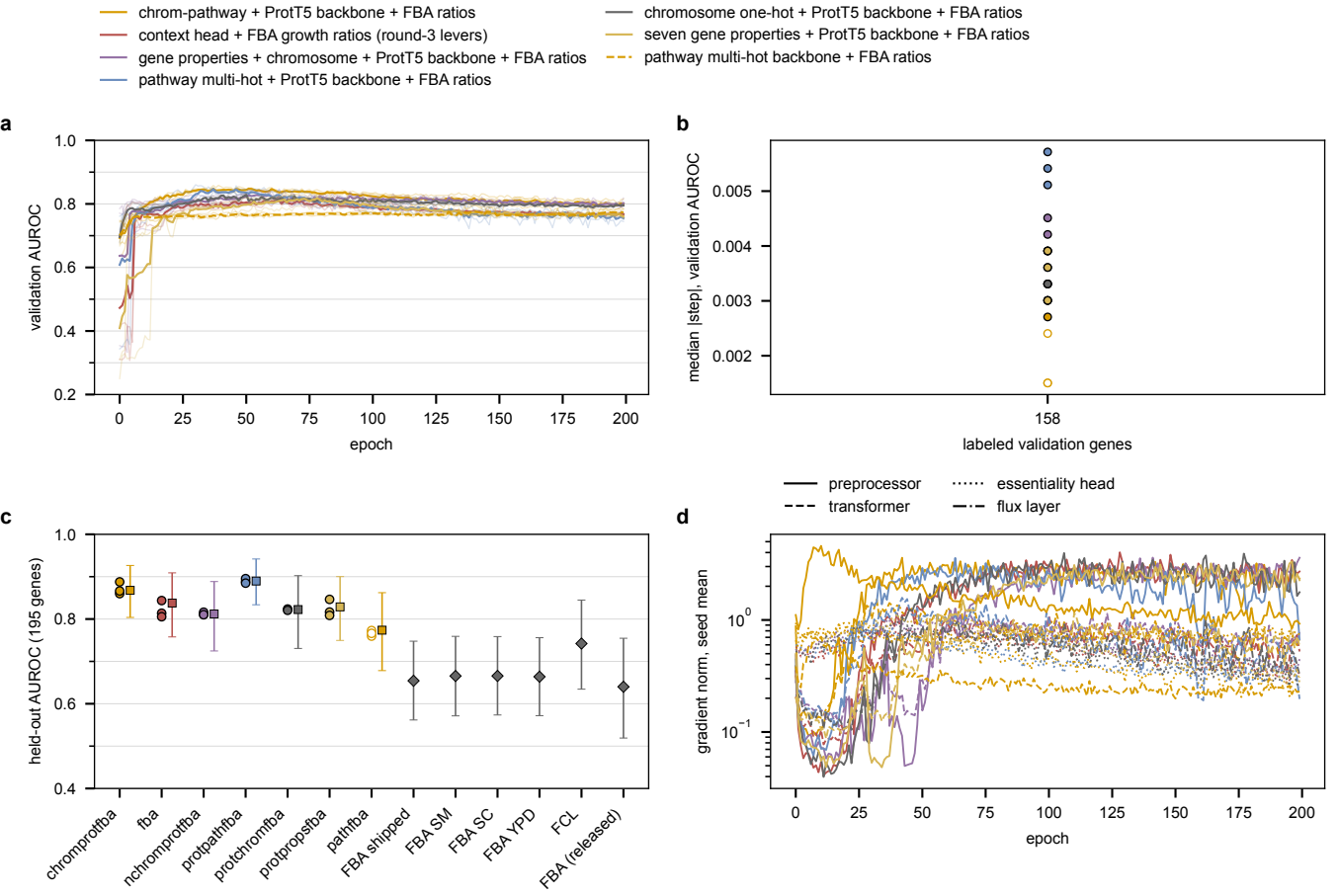

chrom-pathway + ProtT5 backbone + FBA ratios
context head + FBA growth ratios (round-3 levers)
gene properties + chromosome + ProtT5 backbone + FBA ratios
pathway multi-hot + ProtT5 backbone + FBA ratios
chromosome one-hot + ProtT5 backbone + FBA ratios
seven gene properties + ProtT5 backbone + FBA ratios
pathway multi-hot backbone + FBA ratios
a
validation AUROC
epoch
b
median |step|, validation AUROC
158
labeled validation genes
preprocessor
transformer
essentiality head
flux layer
c
held-out AUROC (195 genes)
chromprotfba
fba
nchromprotfba
protpathfba
protchromfba
protpropsfba
pathfba
FBA shipped
FBA SM
FBA SC
FBA YPD
FCL
FBA (released)
d
gradient norm, seed mean
epoch